

Opportunities Hub

7-week workflow

A platform connecting Cambodian students to internships, jobs, scholarships, volunteer work, and competitions. Three members. Seven weeks. One launch.

Project	Opportunities Hub — opportunity discovery platform
Tech stack	React · Node.js · Express · PostgreSQL · Docker
Duration	7 weeks · 3 team members
Courses	Backend Dev · DB Admin · SE · Automata · HCI · Research Methodology

THE TEAM

M1	Member 1 · Backend & Database API · DB schema · server logic	M2	Member 2 · Frontend & UX UI · React · usability testing	M3	Member 3 · Research & Docs UML · report · presentation
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OVERVIEW · 7 WEEKS

WEEK	PHASE	FOCUS	KEY OUTPUT
01	DISCOVERY	Research & Planning	Project proposal · survey results · tech stack dec...
02	DESIGN	System & UI Design	ER diagram · UML diagrams · Figma mockups · API sp...
03	FOUNDATION	Database & Setup	Working database · React skeleton · Express server...
04	CORE BUILD	Core Features (MVP)	MVP working — users can browse, search, and view o...
05	EXTEND	User Roles & Features	3-role system live — Student, Organization, Admin ...
06	POLISH	Testing & Refinement	Stable, tested app with usability report & bug log
07	LAUNCH	Deploy & Present	Live deployed app + full report + presentation + d...

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of 07

Research & Planning

"Understand the problem before solving it."

Research Methodology	Software Engineering	HCI
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M1 · BACKEND & DB • Research similar platforms (LinkedIn, Indeed, Khmer24) • List required data entities (users, posts, categories) • Choose tech stack (Node + PostgreSQL)	M2 · FRONTEND & UX • Survey 20+ students (Google Form) • Define user personas (student / organization) • Collect design inspiration (Dribbble, Behance)	M3 · RESEARCH & DOCS • Write project proposal • Define scope, problem statement, objectives • Set up GitHub repo & project board
DELIVERABLE	Project proposal · survey results · tech stack decision · GitHub repo	

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System & UI Design

"Draw it before you build it."

Software Engineering	Database Admin	HCI	Automata
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M1 · BACKEND & DB • Design ER diagram (users, opportunities, applications) • Normalize tables to 3NF • Plan API endpoints (REST routes) • State diagram: opportunity status flow	M2 · FRONTEND & UX • Build wireframes in Figma • Design color palette & typography • Create high-fidelity mockups (home, listing, detail)	M3 · RESEARCH & DOCS • Create UML use case diagram • Class diagram & sequence diagrams • Write functional requirements document	
DELIVERABLE	ER diagram · UML diagrams · Figma mockups · API specification		

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Database & Setup

"Lay the foundation everyone will build on."

Database Admin	Backend Dev	HCI
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M1 · BACKEND & DB • Set up PostgreSQL database • Create tables, indexes, foreign keys • Seed sample data (10+ opportunities) • Initialize Express server	M2 · FRONTEND & UX • Initialize React app (Vite) • Build layout components (navbar, footer) • Set up routing & pages • Design system: buttons, cards, inputs	M3 · RESEARCH & DOCS • Write database documentation • Document API endpoints (Swagger / Postman) • Conduct usability pre-tests on mockups
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DELIVERABLE	Working database · React skeleton · Express server running locally
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Core Features (MVP)

"Make the main flow work end-to-end."

Backend Dev	Database Admin	Automata	HCI
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M1 · BACKEND & DB • Build API: list / create / update opportunities • Implement search & filter endpoints • Input validation with regex (automata) • User authentication (JWT tokens)	M2 · FRONTEND & UX • Build homepage with opportunity list • Search bar & filter sidebar • Detail page for each opportunity • Connect to backend API	M3 · RESEARCH & DOCS • Start writing project report (intro, methodology) • Track sprint progress on GitHub board • Test build manually, log bugs
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DELIVERABLE	MVP working — users can browse, search, and view opportunities
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User Roles & Features

"Two sides, one platform."

Backend Dev

HCI

Software
Engineering

M1 · BACKEND & DB

- Organization side: post / edit / delete opportunities
- Admin panel: approve / reject postings
- Bookmark & save features for students
- Deadline state automation

M2 · FRONTEND & UX

- Organization dashboard UI
- Admin moderation interface
- User profile page & saved items
- Make mobile-responsive

M3 · RESEARCH & DOCS

- Write system design chapter of report
- Update UML with any design changes
- Plan user test sessions for Week 6

DELIVERABLE

3-role system live — Student, Organization, Admin all working

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Testing & Refinement

"Find bugs before your audience does."

HCI	Software Engineering	Research Methodology
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M1 · BACKEND & DB • Fix all critical & high bugs • Optimize slow queries & add indexes • Add error handling & logging • Backup & recovery strategy	M2 · FRONTEND & UX • Run usability tests with 5+ students • Apply feedback & polish UI • Add loading states & empty states • Improve accessibility (alt text, contrast)	M3 · RESEARCH & DOCS • Compile usability test results • Write evaluation & results chapter • Take screenshots for report & slides
DELIVERABLE	Stable, tested app with usability report & bug log	

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Deploy & Present

"Ship it and tell the story."

All 6 courses

M1 · BACKEND & DB

- Deploy backend to Render
- Deploy database (Railway / Supabase)
- Set up environment variables & security
- Final smoke testing

M2 · FRONTEND & UX

- Deploy frontend to Vercel
- Final UI polish & animations
- Record demo video
- Prepare demo flow for presentation

M3 · RESEARCH & DOCS

- Finalize project report (all chapters)
- Design presentation slides
- Practice presentation 3+ times
- Submit deliverables

FINAL OUTPUT

Live deployed app + full report + presentation + demo video